

Proposal: Next-Generation Vertical Mobility & Cabin Environment Optimization

Executive Summary

Modern high-rise infrastructure demands a shift from **reactive vertical transport** to **predictive logistics**. Traditional "up/down" elevator systems create frustrating wait times, passenger clustering, and inefficient energy consumption. Furthermore, enclosed cabin ventilation remains a secondary thought rather than an active health safety feature. This proposal outlines an integrated framework to transform building transit using Destination Control Systems (DCS), AI Predictive Pre-Positioning, Smart Load Management, and Active Air Purification.

1. Core Objectives

- **Minimize Total Journey Time:** Reduce average passenger wait and transit times by up to 40% through intelligent passenger grouping.
- **Optimize Space & Load Allocation:** Prevent overcrowded "FULL" cabins from making unnecessary, frustrating stops using advanced weight sensor integrations.
- **Sanitize the Cabin Environment:** Implement continuous, medical-grade air purification to eliminate airborne pathogens and stale shaft air.
- **Reduce Energy Consumption:** Utilize machine learning to transition idle elevator fleets into energy-saving modes without sacrificing readiness during peak demand.

2. Proposed Technical Framework

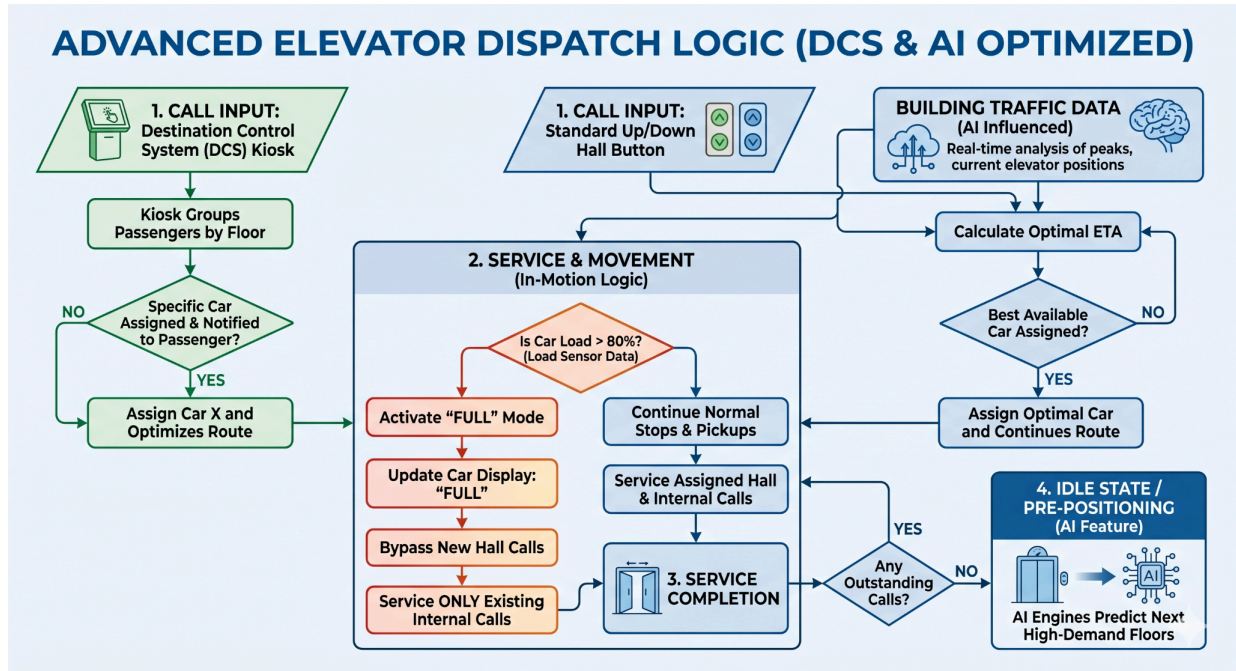
Phase A: Intelligent Traffic Dispatch (The Logic)

Instead of traditional floor buttons, the building will transition to a **Destination Control System (DCS)** combined with a real-time AI dispatch engine.

- **Lobby Kiosk Grouping (DCS):** Passengers enter their destination floor *before* boarding. The system instantly groups travelers heading to the same floors into the same car, turning local trips into optimized express routes.
- **AI Predictive Pre-Positioning:** The system analyzes historical building traffic patterns to anticipate spikes (e.g., morning rushes, lunch hours). Idle cars are autonomously "parked"

at high-demand zones *before* a button is ever pressed.

- **Smart "FULL" Bypassing:** When internal load-sensors detect cabin capacity exceeding **80%**, the car automatically triggers **"FULL" Mode**. It updates the external display and bypasses all external hall calls, focusing strictly on dropping off current passengers safely and quickly.



Phase B: Advanced Cabin Air Management (The Environment)

To solve the "aerosol trap" and "shaft air" issues common in high-density buildings, the ventilation system will be modernized into a self-contained, protective biosphere.

- **Active UV-C & Bipolar Ionization:** Integrated ceiling plenums pull air through a shielded high-intensity UV-C chamber to destroy viral/bacterial DNA, while releasing safe ions to clump and drop dust and micro-allergens out of the breathing zone.
- **Positive Pressure Architecture:** Fans will maintain higher air pressure inside the cabin relative to the elevator shaft. When doors open, air rushes *out* into the lobby, completely preventing dusty shaft air or stagnant hallway air from entering the car.
- **Dynamic Purge Cycles:** Using weight sensors, the ventilation system automatically scales up cubic feet per minute (CFM) during high-occupancy trips and runs an accelerated air flush cycle immediately after a crowd exits.

3. Expected Business Impact & ROI

Metric	Traditional System	Proposed Smart System	Net Impact
Avg. Lobby Wait Time	High (Sequential Stops)	Minimal (Pre-Positioned / Grouped)	~35-50% Reduction
Mid-Trip Stops per Journey	Frequent (Stops at every hall call)	Optimized (Direct/Express Grouping)	~40% Fewer Stops
Cabin Air Quality	Low (Recirculated shaft dust)	Pristine (UV-C filtered, positive pressure)	99.9% Pathogen Elimination
System Energy Wear	Continuous 24/7 Peak Strain	Dynamic Smart-Scaling & Standby	~15-25% Energy Savings

4. Implementation Next Steps

- Traffic Diagnostics:** Audit the building's current peak-hour data, existing floor utilization rates, and average passenger payload statistics.
- Hardware Retrofitting:** Install DCS kiosks in main lobbies and integrate load-weight sensors and UV-C filtration modules into existing cabins.
- Software Deployment:** Layer the AI predictive dispatch algorithm over the existing motor controllers to begin the continuous learning and mapping cycle.
- Testing & Calibration:** Run off-peak simulations to adjust "FULL" weight thresholds and test the positive pressure ventilation curves.

This document was developed with the assistance of AI.

Initial Idea:

Have to wait long time (impatient)

-> mirrors / digital infotainment screens next to the elevators

Ventilation system (power consumption)

-> dynamic, occupancy-based ventilation (IoT)

-> display FULL when packed (in that case maybe only going to user-pressed floor)

* going up and down to respective floor with less amount of waiting time